TEXORA · LECTURE SHEET

Machine Design

From analysis to decision: the design process, allowable stress and factor of safety, stress concentration at shoulders and keyways, fatigue and the endurance limit — then two complete element designs, a shaft under combined bending and torsion and an open belt drive.

ME-3210 · SOLID MECHANICS AND MACHINE DESIGN · TOPIC 5 · LEVEL 3 TERM 2

1. The design process and allowable stress

Design is analysis run backwards with judgement: define the function and loads, choose the material, propose geometry, calculate stresses, compare with an **allowable stress**, iterate until the comparison passes *and* the part is stiff enough, stable, manufacturable and affordable. The allowable comes from a strength divided by a **factor of safety** chosen for the uncertainty and the consequences: yield strength for ductile static design (FoS ≈ 1.5–2 on yield), ultimate for brittle materials (FoS 3–4), fatigue strength for cyclic duty. Codes compress this experience into permitted stresses — e.g. the ASME shaft code's 0.75 factor on keyed shafts, which we spend below.

2. Stress concentration

Wherever section changes abruptly — shoulder fillet, keyway, hole, thread root, circlip groove — stress multiplies locally by the **stress concentration factor** K (from charts against geometry ratios): $\sigma_{max} = K \times \sigma_{nominal}$. A perfectly sharp corner makes K theoretically unbounded, which is why real parts crack at shoulders and why fillet radii are specified, not suggested. Under static ductile loading local yielding blunts K's effect; under fatigue (next section) it applies with almost full cruelty, softened only slightly by the material's notch sensitivity.

3. Fatigue — failure below the yield stress

Most machine failures are fatigue failures: repeated cycles grow a crack from a surface defect or notch until the remaining section snaps — at stresses the part survived happily once. Rotating shafts live here: every revolution bends each surface fibre from tension to compression and back. Test data plot as the **S–N curve** (stress amplitude vs cycles to failure): steels show a knee at the **endurance limit** $\sigma_e \approx 0.5 \times$ UTS (rotating-beam, polished specimen), below which life is effectively infinite; aluminium alloys show no knee and are rated at a stated life (e.g. 5×10^8 cycles).

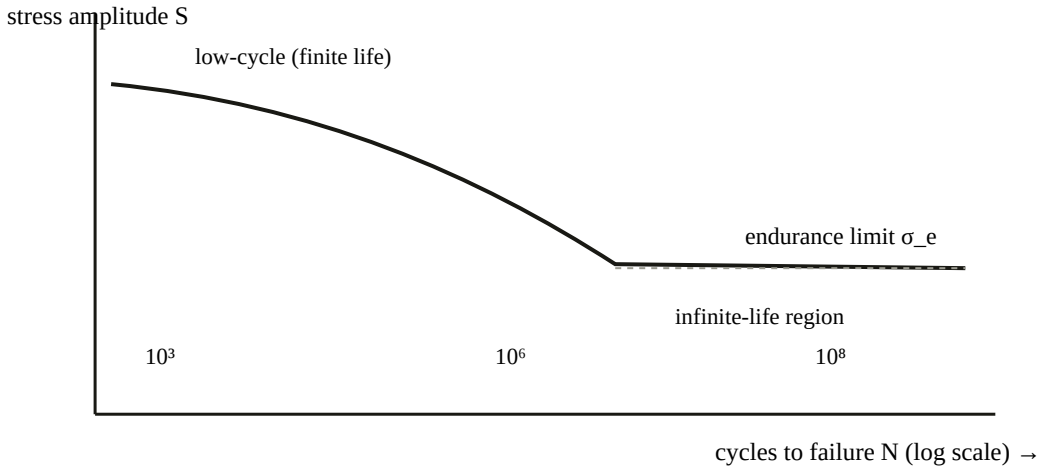


Figure 1 — S–N curve for steel: falling finite-life branch, then the knee into the horizontal endurance limit. Design for rotating parts aims below σ_e — after derating it.

The laboratory limit must be **derated** for real life: surface finish, size, reliability, stress concentration, corrosion and temperature each take their factor (Marin factors), commonly halving the usable endurance strength of a machined shaft. Mean stress matters too: a steady torque (mean shear) superimposed on alternating bending is judged on a **Soderberg/Goodman diagram** — allowable amplitude shrinks as mean stress grows, with yield as the outer wall. Design rule of thumb: keep the alternating stress, after all factors, safely under the endurance line, and remember a loom shaft may accumulate 10^8 cycles within a few years.

4. Worked design — a shaft in bending and torsion

WORKED EXAMPLE — COMBINED LOADING (ASME APPROACH)

A shaft carries a steady torque T = 400 N·m and a bending moment M = 300 N·m at the critical section (a gear seat). Equivalent twisting moment: $T_e = \sqrt{(M^2 + T^2)} = \sqrt{(300^2 + 400^2)} =$ **500 N·m** (the 3-4-5 triangle is not a coincidence of nature — check your arithmetic when it appears). Material allowance with keyway (ASME): $\tau_{allow} = 42$ MPa. From $\tau = 16T_e/(\pi d^3)$: $d^3 = 16 \times 500/(\pi \times 42 \times 10^6) = 6.06 \times 10^{-5}$ m³ → d = 39.3 mm → **specify 40 mm**. Remaining checks before release: deflection at the gear (mesh misalignment), critical speed, and the key itself (shear and crushing) — strength alone never finishes a shaft design.

5. Worked design — open flat belt drive

A belt transmits power by friction: the tight side tension T_1 exceeds the slack side T_2, and at impending slip the **capstan (belt friction) relation** caps the ratio:

$$T_1/T_2 = e^{(\mu\theta)}, \quad P = (T_1 - T_2)\, v$$

WORKED EXAMPLE — BELT POWER

Leather belt on a cast-iron pulley, $\mu = 0.3$, lap angle $\theta = 180° = \pi$ rad, belt speed 20 m/s, maximum permitted tension $T_1 = 1000$ N. Ratio $= e^{(0.3\pi)} = e^{0.9425} = 2.566$, so $T_2 = 1000/2.566 =$ **390 N**. Power $= (1000 - 390) \times 20 =$ **12.2 kW**. Note the levers: power grows with speed and with the tension *difference*; more lap (larger pulley centre distance/geometry or an idler) raises the slip ceiling exponentially. Centrifugal tension at high speed ($T_c = mv^2$) eats the margin — one reason belts yield to chains and gears at high power density, yet survive everywhere for their cheapness, quietness and slip-as-a-fuse behaviour.

6. Key points

- Allowable stress = strength ÷ factor of safety, the factor priced by uncertainty and consequence; codes hand you pre-priced values.
- Shoulders, keyways and holes multiply local stress by K; fillets are functional, fatigue applies K almost fully.
- Fatigue rules rotating machinery: steels have an endurance limit ≈ 0.5 UTS (specimen), derated hard for real surfaces and notches.
- Combined M + T: use $T_e = \sqrt{(M^2 + T^2)}$ → the example shaft sizes to 40 mm at 42 MPa.
- Belts: $T_1/T_2 = e^{(\mu\theta)}$, $P = (T_1 - T_2)v$ — 12.2 kW for the example drive.

7. Practice

Answer in your own words; brief model answers follow.

1. A shaft carries M = 500 N·m and T = 500 N·m; τ_{allow} = 40 MPa. Size the solid shaft.
2. A steel has UTS 600 MPa. Estimate its rotating-beam endurance limit, then the usable design value if combined derating factors total 0.55.
3. A belt drive has $\mu = 0.25$ and lap 150°. Find the tension ratio. If $T_1 = 1200$ N and $v = 15$ m/s, find the power.
4. Two otherwise identical shafts differ only in shoulder fillet radius: 1 mm vs 4 mm. Which fails first under rotating bending, and through which two mechanisms does the radius act?
5. A brittle cast-iron bracket and a ductile steel bracket carry the same static load. Compare design strengths and safety factors you would use, and say why they differ.
6. Why is “make it twice as thick” not a universal cure for fatigue failure?

Answer key

1. $T_e = \sqrt{(500^2 + 500^2)} = 707$ N·m. $d^3 = 16 \times 707/(\pi \times 40 \times 10^6) = 9.00 \times 10^{-5}$ m³ → d = 44.8 mm → specify 45 mm.
2. $\sigma_e \approx 0.5 \times 600 = 300$ MPa (specimen). Usable = 0.55 × 300 = 165 MPa before any further safety factor — the gap between 600 MPa UTS and ~165 MPa working alternating stress is the honest price of fatigue design.
3. $\theta = 150° = 2.618$ rad; ratio = e^(0.25 × 2.618) = e^0.6545 = 1.92. $T_2 = 1200/1.92 = 624$ N. P = (1200 − 624) × 15 = 8.64 kW.
4. The 1 mm fillet shaft: its stress concentration factor K is higher (sharper notch), so local alternating stress at the shoulder is higher; and fatigue cracks initiate exactly at that surface notch. The 4 mm fillet lowers K and spreads the stress flow — same nominal stress, longer life.
5. Steel (ductile): design on yield, FoS ≈ 1.5–2, since yielding warns and redistributes local peaks. Cast iron (brittle): design on ultimate strength with FoS ≈ 3–4 — no yield warning, wide scatter, and internal flaws make failure sudden; the extra factor buys the warning the material refuses to give.
6. Fatigue is governed by local stress at notches and by surface condition, not bulk size alone: a thicker part with the same sharp shoulder keeps the same K (and larger size slightly raises the statistical flaw population and can lower the endurance factor). Cure the notch, the finish and the mean stress — not just the section.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib